

Algorithmic Precision in Skin Lesion Diagnosis



The Stakes:

Melanoma is highly curable if caught early, but deadly if missed.

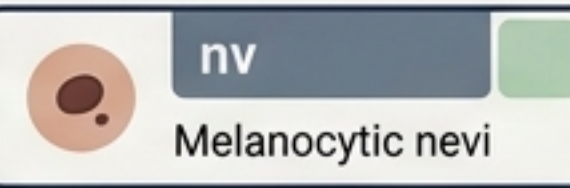
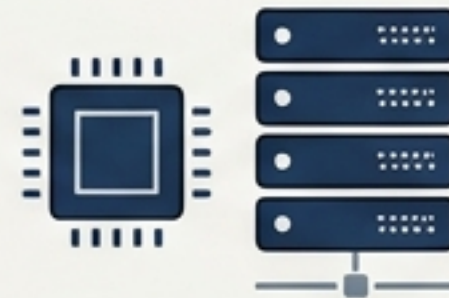
Human interpretation is subjective.

Mistakes cost lives.



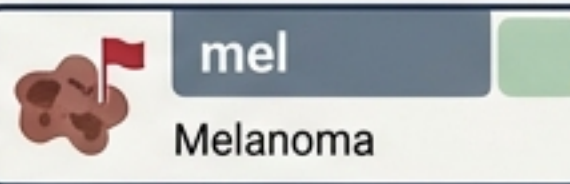
Input: 1 Dermatoscopic Image (600 × 450px)

The Classification Engine



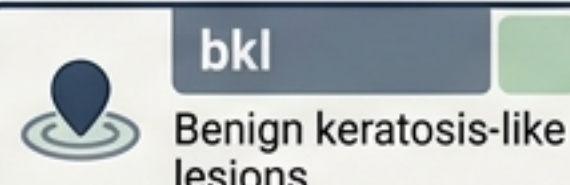
nv

Melanocytic nevi



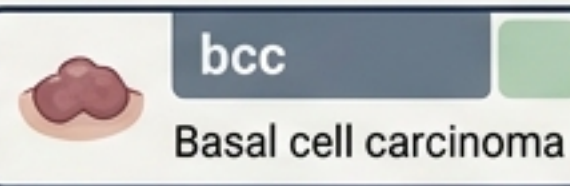
mel

Melanoma



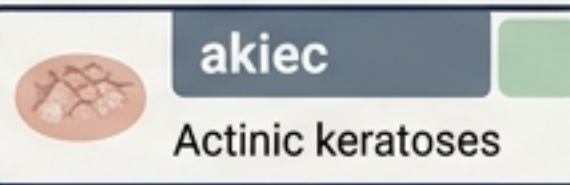
bkl

Benign keratosis-like lesions



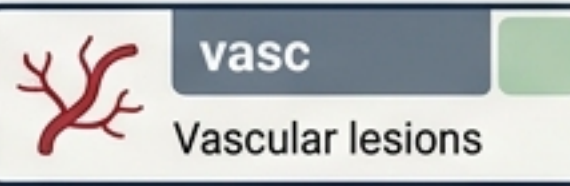
bcc

Basal cell carcinoma



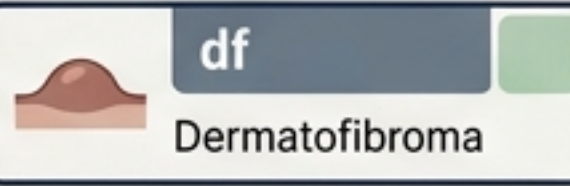
akiec

Actinic keratoses



vasc

Vascular lesions



df

Dermatofibroma



Strict Computational Boundaries:

- 7-Class Image Classification
- **NOT** Image Segmentation
- **NOT** Binary (Cancer vs. Non-Cancer)

The Two Traps of HAM10000

Dataset Profile: 10,015 photos | 7 classes | 50%+ biopsy-verified

Trap 1: The Imbalance & Lying Metrics



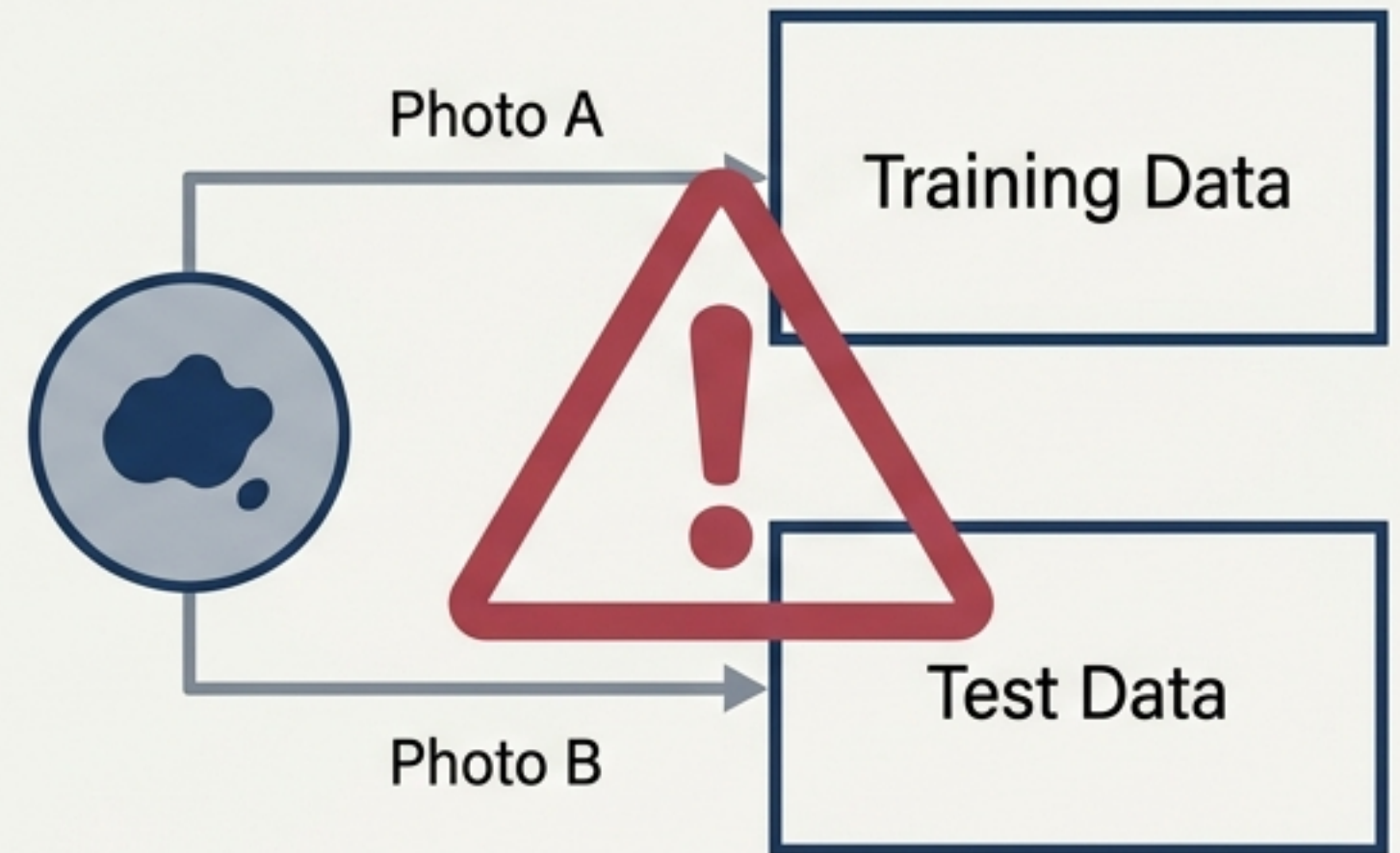
Accuracy



Sensitivity &
Macro AUC

Rule: Discard raw accuracy. We must report Sensitivity (missing cancer is the fatal error) and Macro AUC to give rare classes equal weight.

Trap 2: The lesion_id Leakage



Rule: Splitting randomly by photo puts the same mole in both sets, allowing the model to cheat on the exam. We must strictly split by lesion_id.

Group A: Proof of Concept



Methodology Card

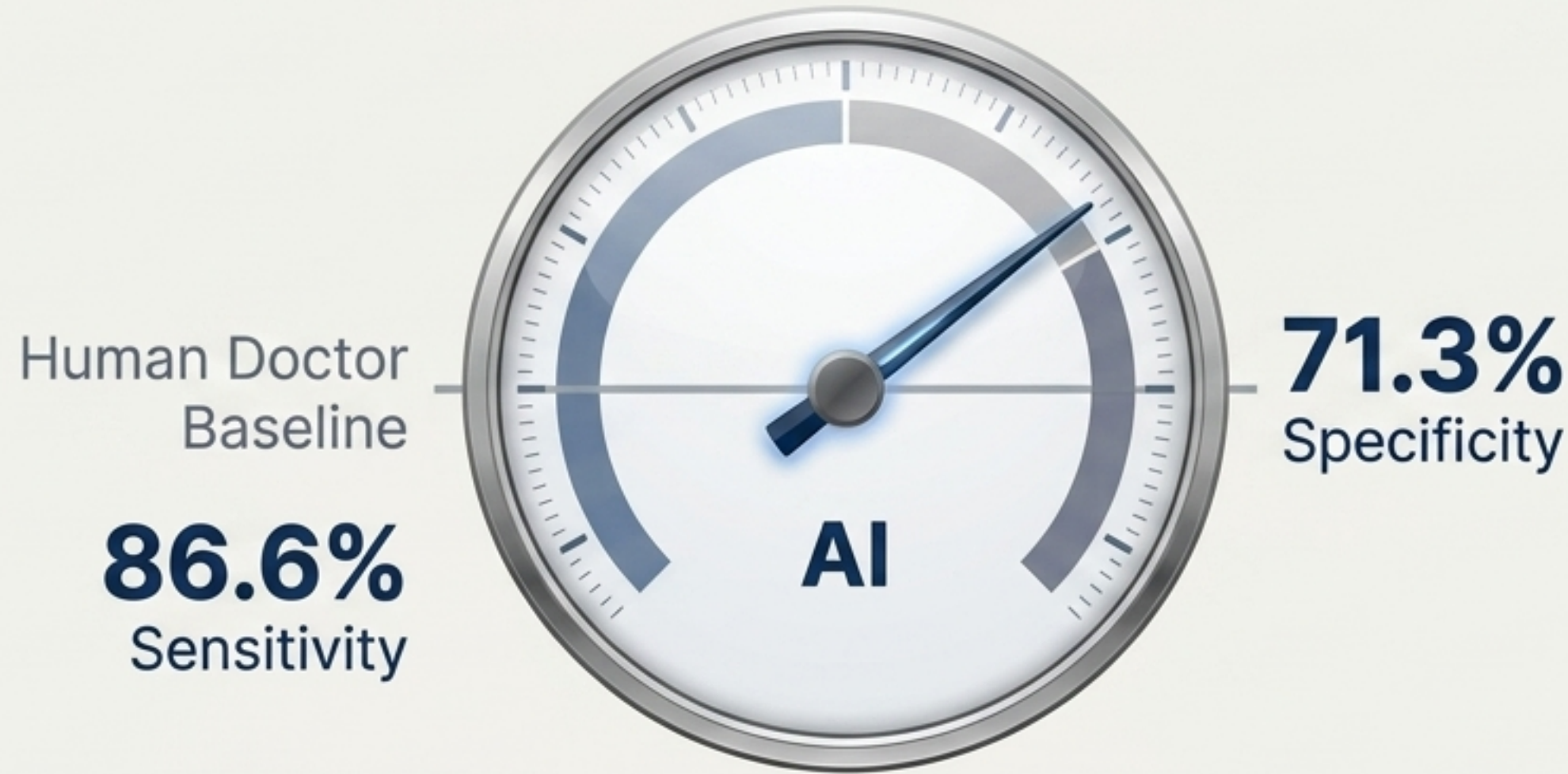
Esteva et al., 2017 (Nature)

- The genesis of computational dermatology.
- Trained on a massive proprietary dataset (not HAM10000).
- Performance matched 21 board-certified dermatologists.

Why It's In Our Report: The Genesis Point.

It proves the core idea is viable and birthed the field, validating the premise of this project. However, the reliance on closed data highlights the need for public datasets.

Group A: Man Against Machine



Haenssle et al., 2018
(Annals of Oncology)

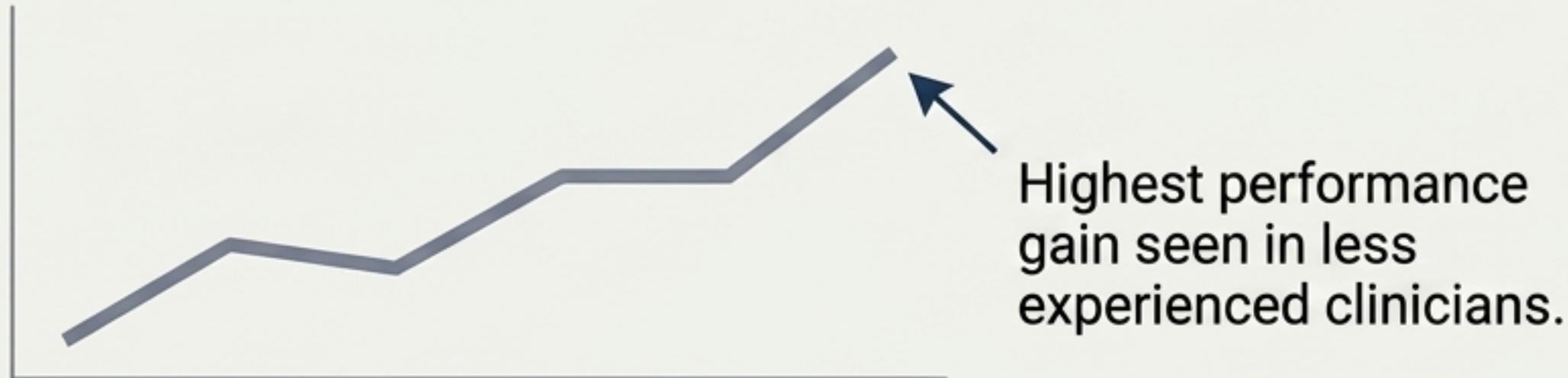
- AI tested against 58 dermatologists from 17 countries.
- AI edged past human specificity.

Why It's In Our Report: The Anchor Metric.

This provides our mandatory human benchmark. Our future model metrics have no meaning unless anchored against this 86.6% real-world human sensitivity.

Group A: The Collaborative Future

Doctor + AI > Doctor Alone
Doctor + AI > AI Alone



Methodology Card

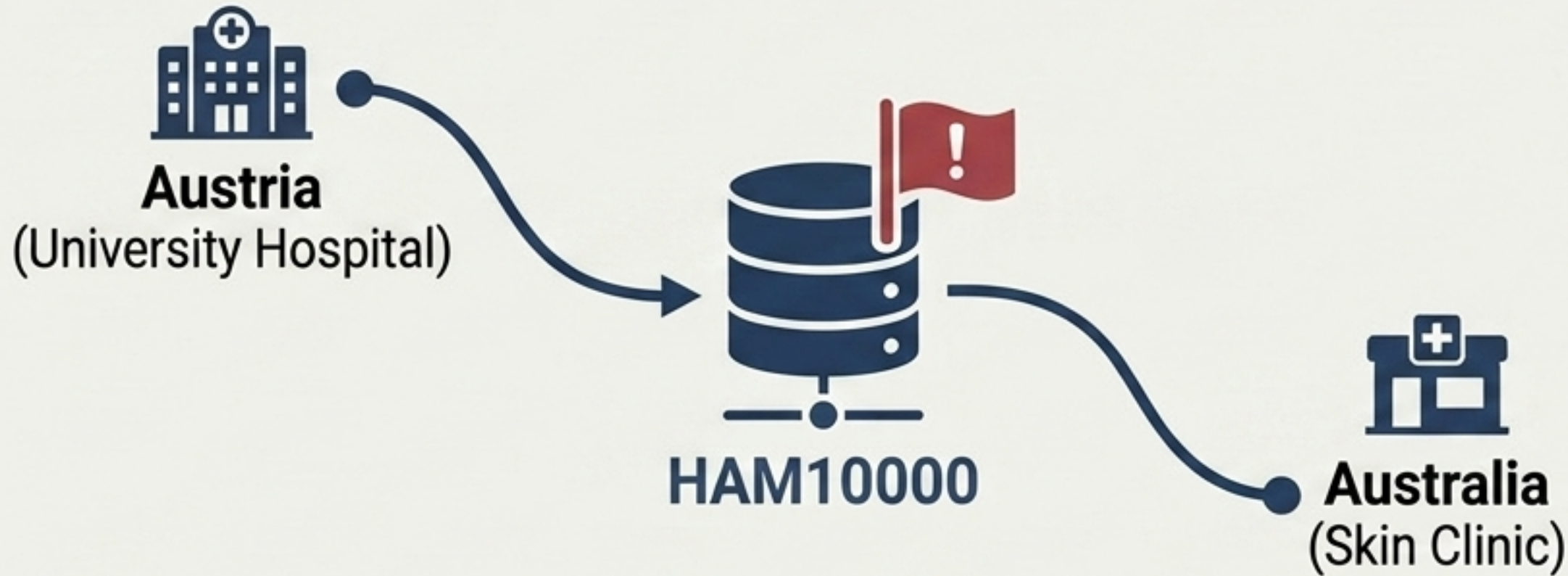
Tschandl et al., 2020
(Nature Medicine)

- Empirical test of human-AI collaboration.
- Proves AI is most effective as a supportive tool, not an autonomous agent.

Why It's In Our Report: The Ethical Boundary.

This is our 'honesty paper.' It strictly informs our project limitations, driving our stance that the model is for decision-support and no autonomous clinical claims will be made.

Group B: The Data Source



Methodology Card

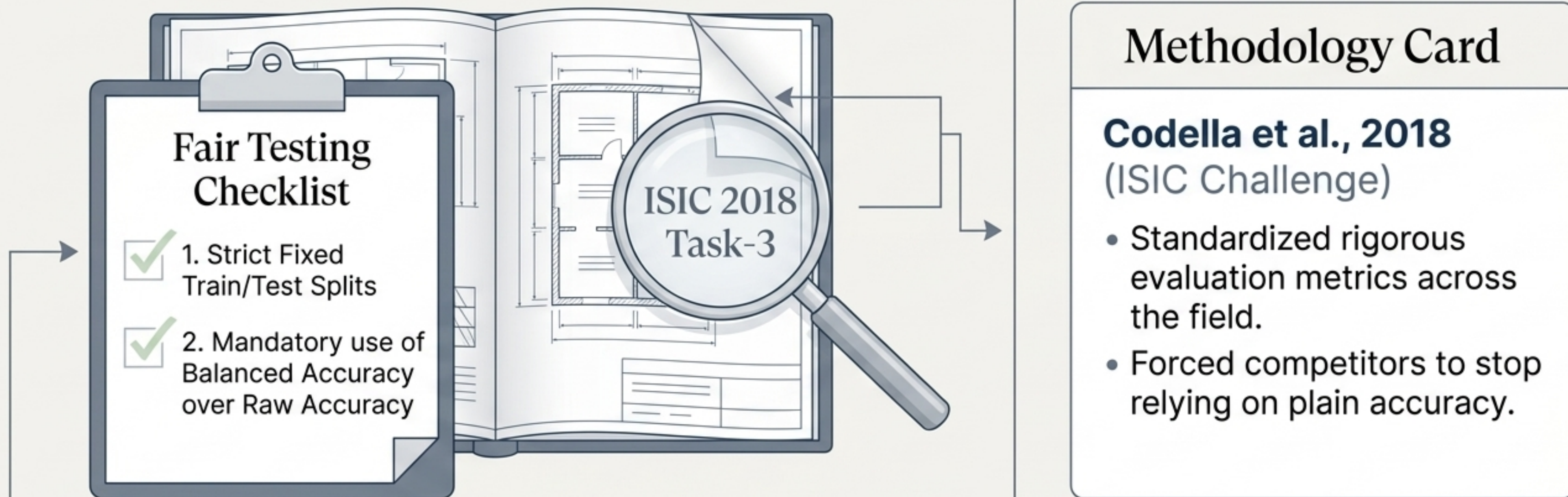
Tschandl, Rosendahl & Kittler, 2018 (Scientific Data)

- The compulsory foundational citation.
- Created and released the 10,015 images.
- Details collection methodology.

Why It's In Our Report: Validating the Leakage Warning.

Crucially, the authors openly admit in the text to capturing multiple photos per mole. This is our definitive proof that the lesion_id leakage trap is a built-in reality of the dataset.

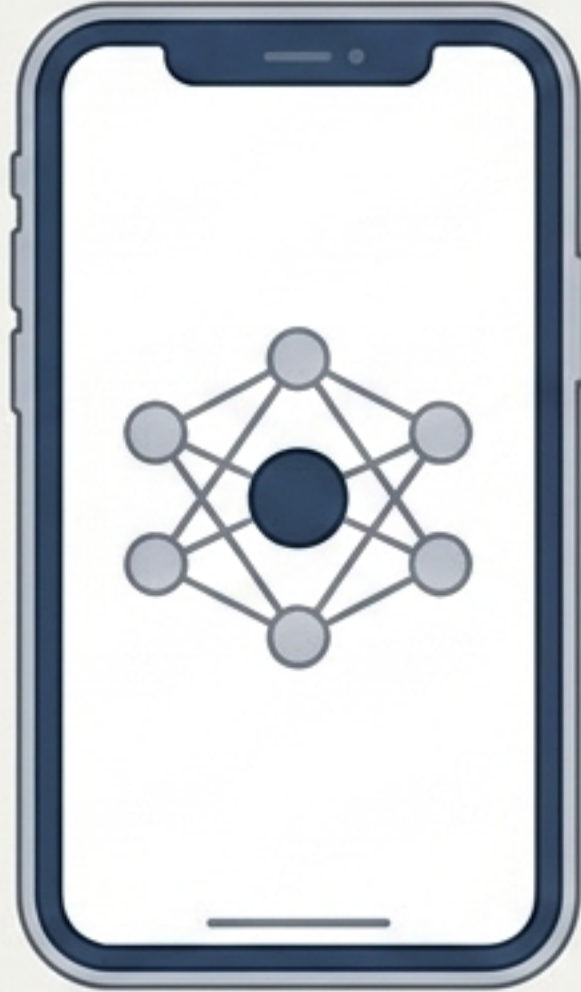
Group B: The Rules of the Game



Why It's In Our Report: The Hidden Alias.

This paper reveals that the "ISIC 2018 Task-3" training set is literally HAM10000. This lets us spot literature that uses our exact data under a different competition alias.

Group C: Simple Models on Our Dataset



Methodology Card

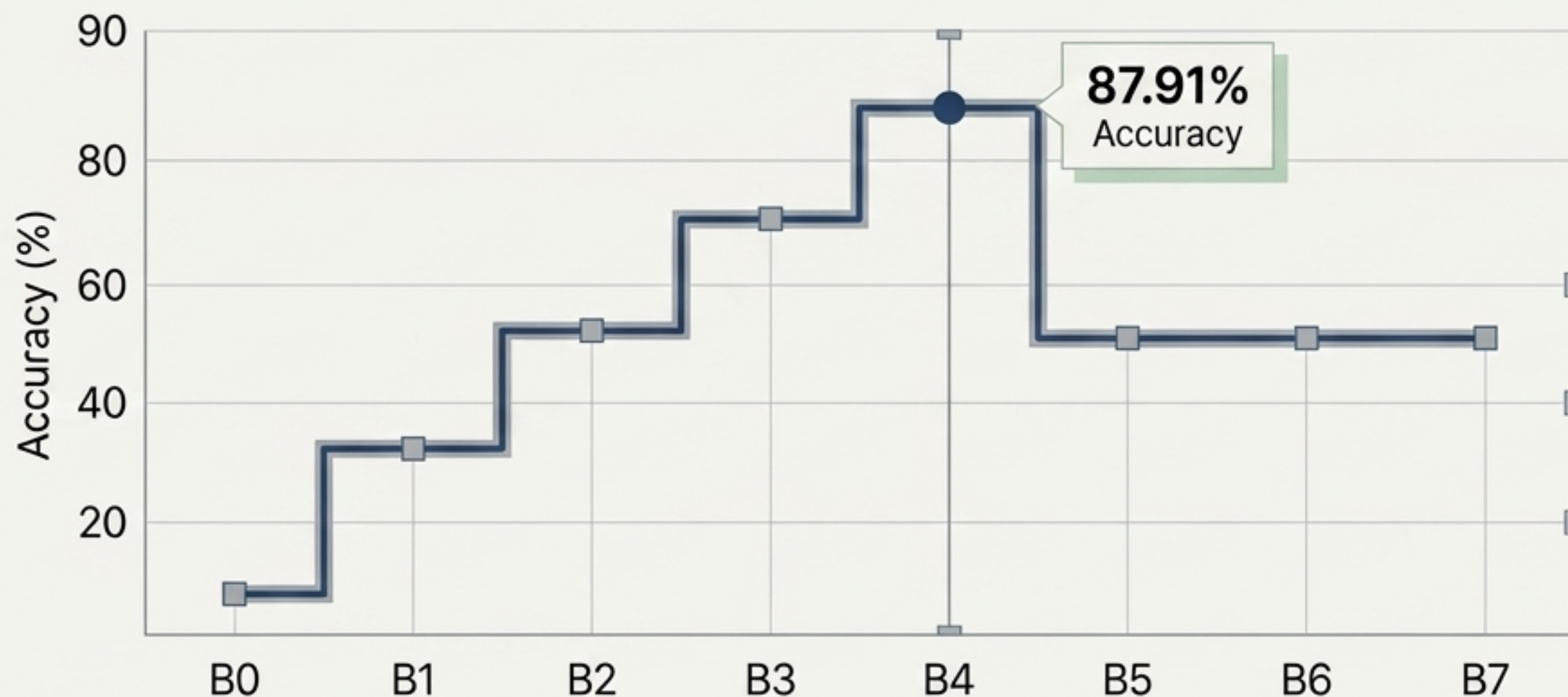
Chaturvedi et al., 2020

- Utilized MobileNet architecture.
- Designed explicitly for low-power mobile devices.
- Fine-tuned directly on HAM10000.
- Achieved 83.1% accuracy.

Why It's In Our Report: The Absolute Floor.

You don't need a supercomputer for baseline results. This is our cheapest honest baseline. Any future, heavier architecture must decisively beat 83.1% to justify its computational cost.

Group C: The Limits of Scaling



Ali et al., 2021

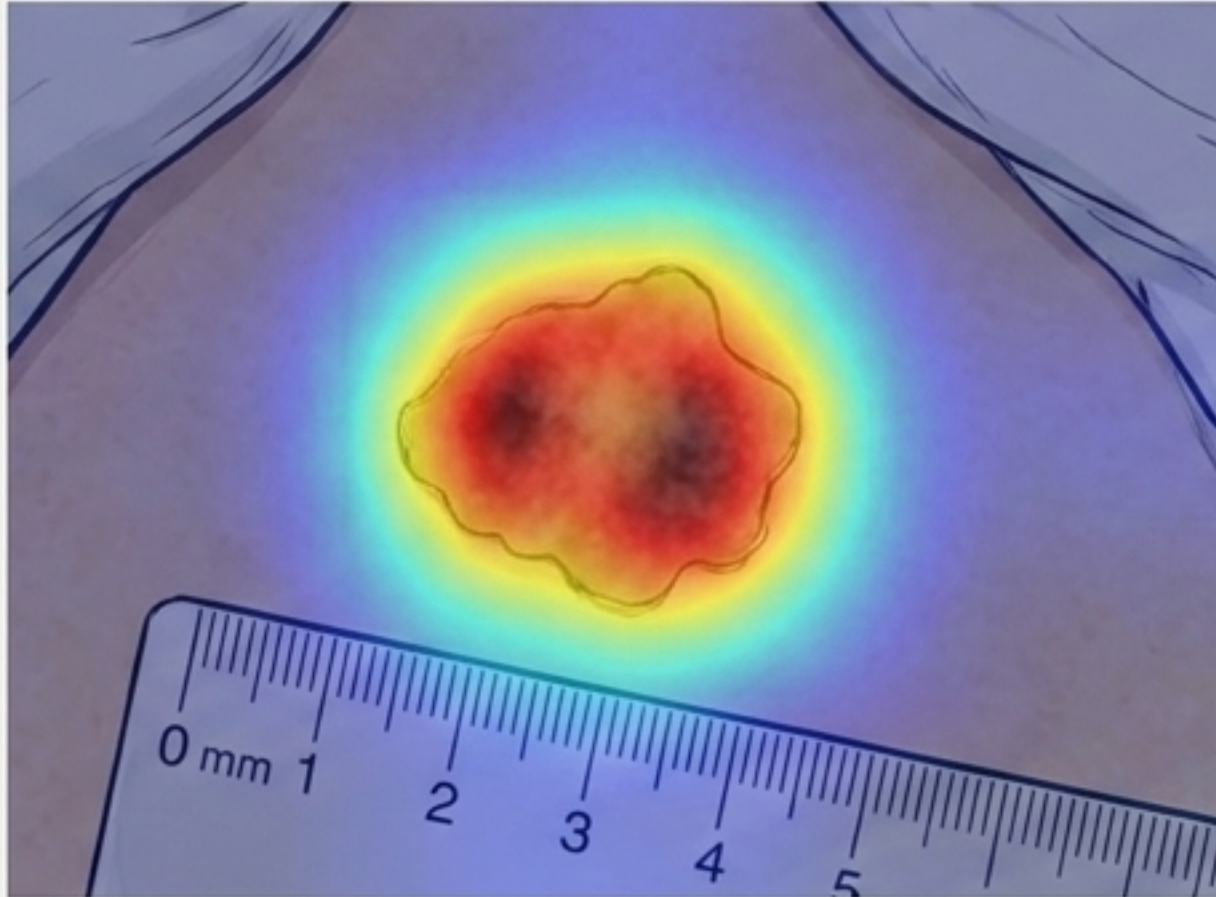
- Tested 8 sizes of the EfficientNet family under identical conditions.
- The massive models (B6, B7) failed to win.
- **Rejected** as base paper: Zero public code released.

Why It's In Our Report: Model Capacity vs. Data Imbalance.

Proves that simply making models larger is a dead end. The core bottleneck with HAM10000 isn't architectural capacity; it is data scarcity and extreme class imbalance.

Group D: Datta et al., 2021 — OUR CHOSEN BASE PAPER

Grad-CAM Localization (Skin Mole)



Methodology Card

- Soft-Attention block added to Inception-ResNet-v2.
- Result: 93.7% precision, 98.4% AUC.
- The attention mechanism added a +3.2% precision bump.

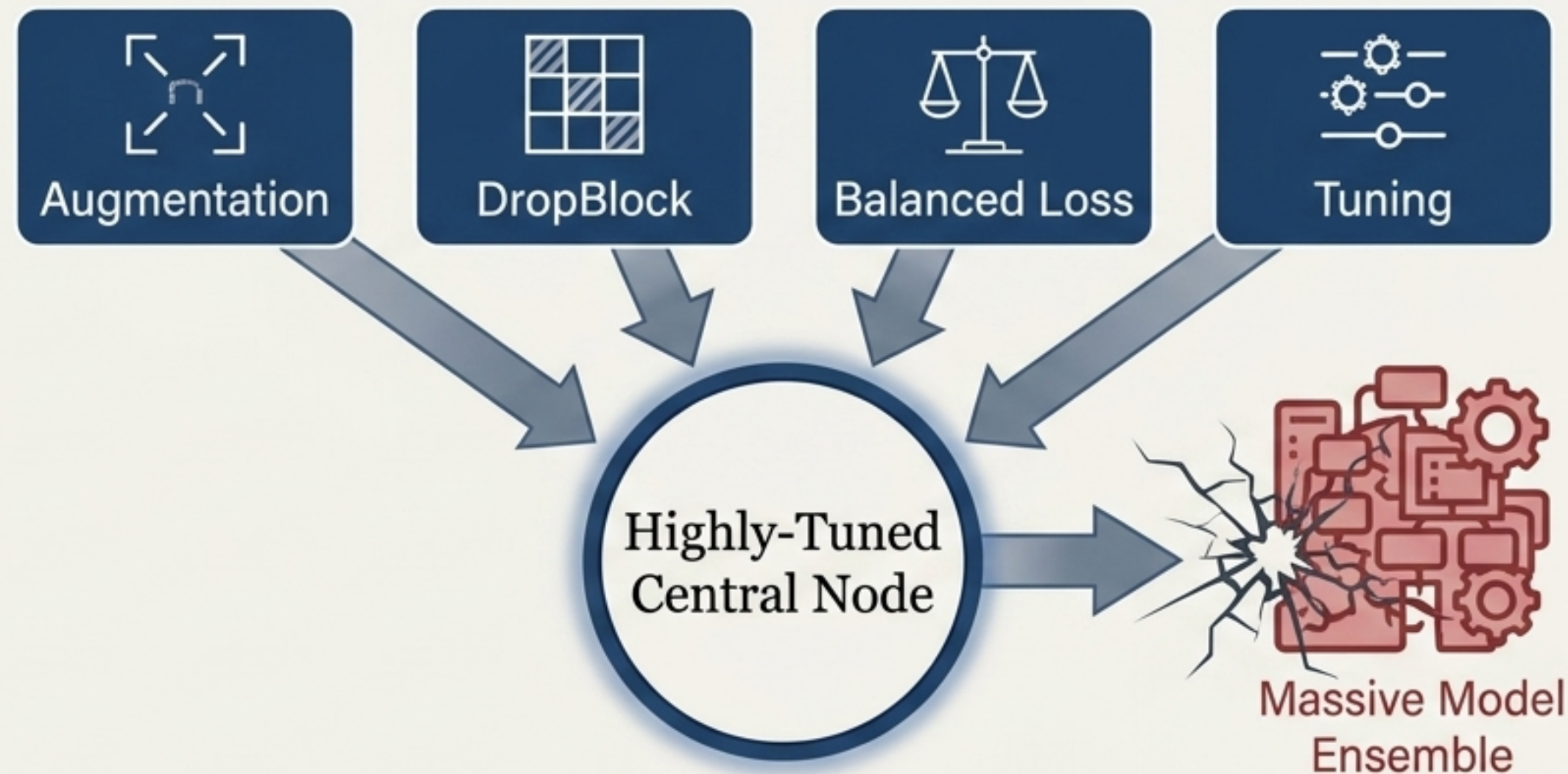
Reproducibility Scorecard: 4.5 / 7

- | | |
|-----------------|--------------------|
| ✓ Public Code | ✓ Hyperparameters |
| ✓ Dataset | ⚠ No Random Seeds |
| ✓ Defined Split | ⚠ No Compute Times |

Why It Won: The Ultimate ‘Before & After’.

They published code for BOTH the plain baselines and the Soft-Attention improvements. We can verify their baseline before testing claims.
Critique: Test set removes multi-photo moles to avoid leakage, shrinking the exam pool to only 828 photos (~10 for rare classes).

Group D: Tackling Imbalance Head-On



Methodology Card

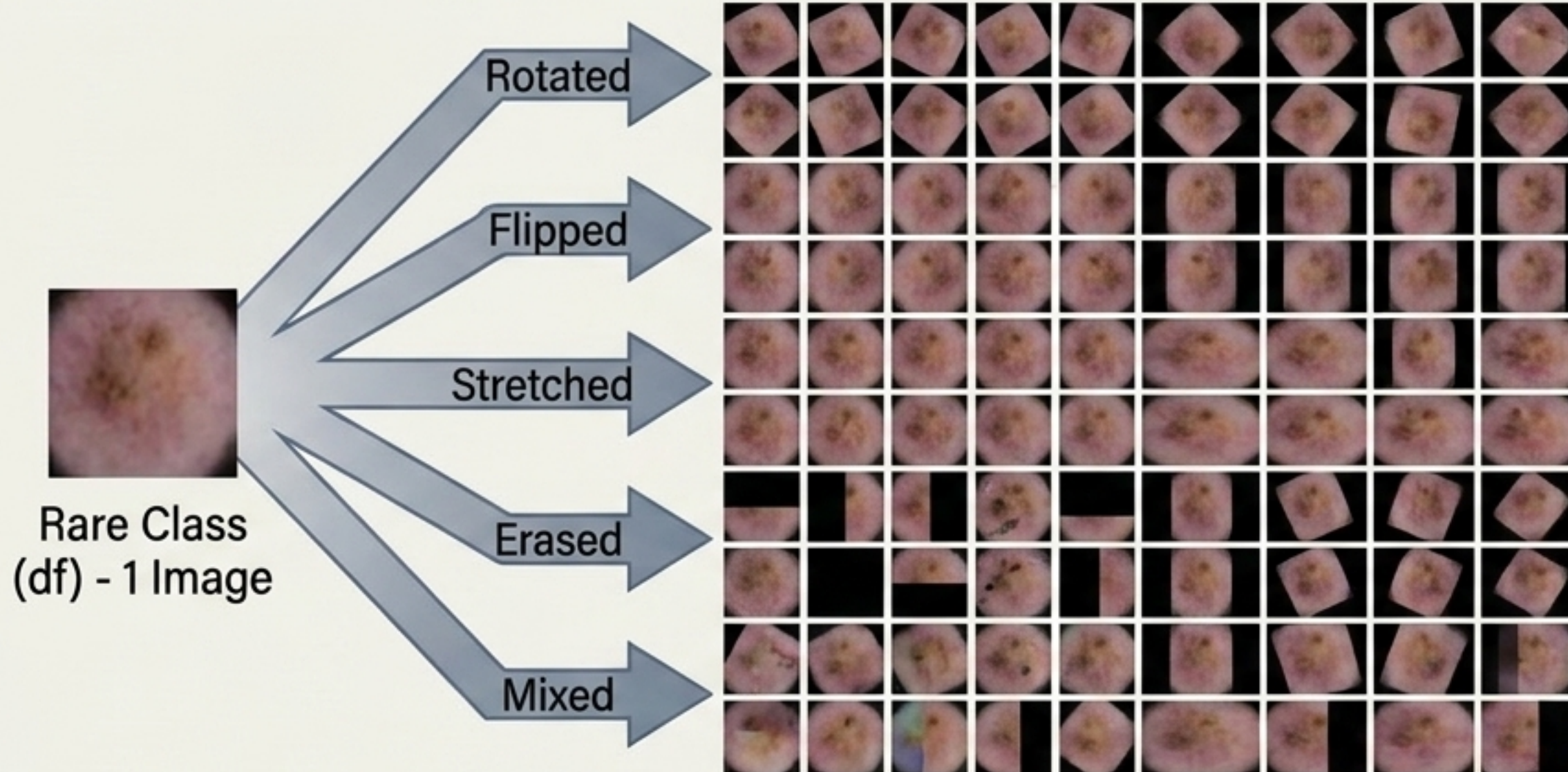
Yao et al., 2022 (IEEE TMI)

- Attacked imbalance head-on with bundled techniques.
- Proved one well-tuned medium model decisively beats a large ensemble.
- Agrees perfectly with the findings of Ali 2021.

Why It's In Our Report: The Runner-Up.

A highly viable alternative path. It featured excellent code quality, but was ultimately rejected as the base paper because it targets ISIC competition configurations over raw HAM10000, and requires GPU overhead beyond our budget.

Group D: The Augmentation Template



Methodology Card

Perez et al., 2018

- When you only have 115 photos of a class, synthetic multiplication is mandatory.
- Tested 13 distinct ways to create artificial training data.
- Methodology: Changed ONE variable at a time to measure exact impact.

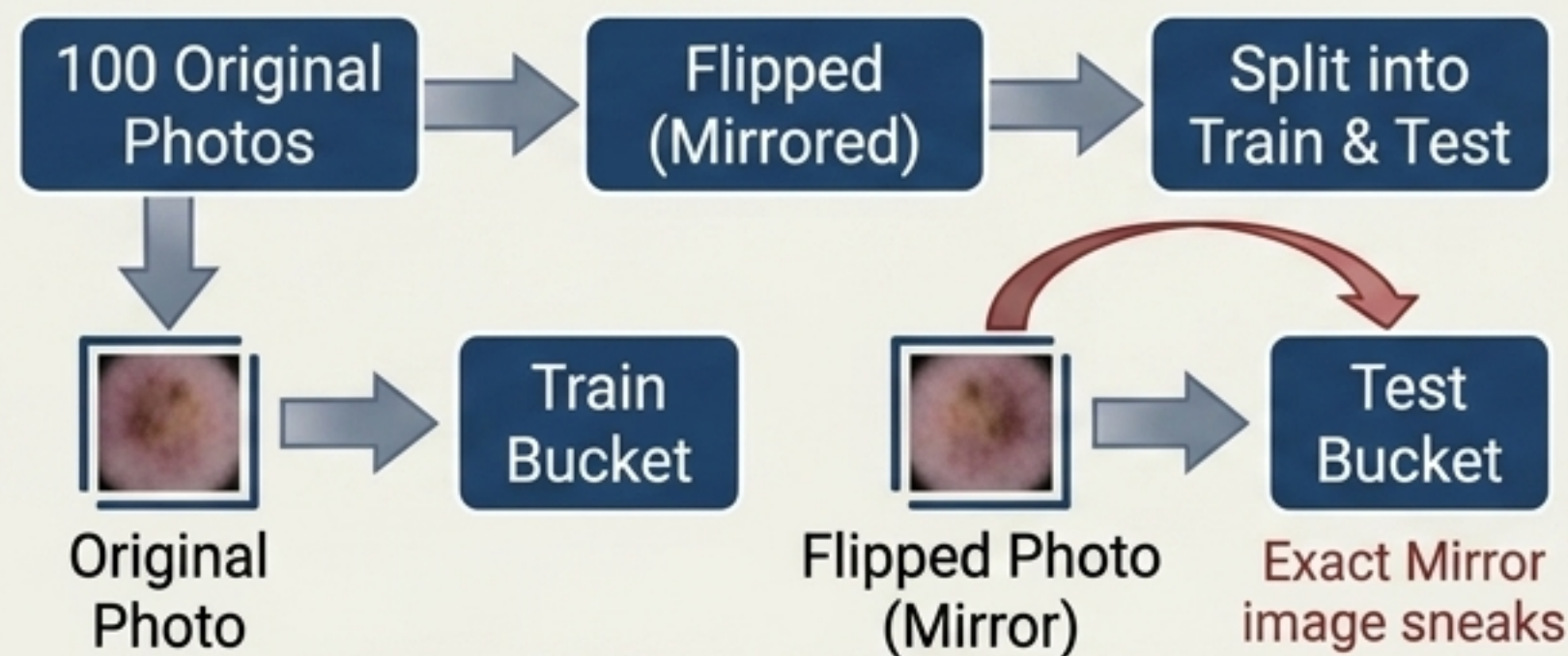
Why It's In Our Report: Our Ablation Blueprint.

This provides the exact methodological template for Milestone 4. When running ablations on our base paper, we will use this strict scientific method: isolate one variable, measure the result, repeat.

Critical Analysis: The Mirage of 95% & Research Integrity

The Flawed Math

~~Shetty et al. (95.18%)~~



Result: Massive leakage

Tested on only 280 photos (2.8% of dataset).
High accuracy is not proof of good methodology.

The Compromised Review



Retracted due to compromised peer review.
Included as proof that we deeply audit literature rather than blindly copying high numbers from reference lists.

Boundaries & Limitations



The Report

This is a survey phase. Reported metrics across papers cannot be perfectly compared as a leaderboard because every author utilized different data splits, preprocessing, and hyperparameter setups.



The Data

HAM10000 has a severe light-skinned demographic bias (sourced from Austria/Australia). Additionally, results for rare classes (df, vasc) will remain statistically shaky due to having only ~100 photos.



The Medicine (**Mandatory**)

We make absolutely NO clinical claims. This is a computational benchmark on a research dataset, not a medical device. It is designed strictly for algorithmic evaluation, never autonomous diagnosis.

The Project Roadmap

